INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

NIGHT VISION SIGHT, CREW SERVED WEAPON AN/TVS-5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Sight, Crew Served Weapon, AN/TVS-5.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to HQ,CERDEC ATTN: RDER-PRO, Aberdeen Proving Ground, MD 21005 or emailed to usarmy.APG.cerdec.mbx.standardization-crx@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government Documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

COMMERCIAL ITEM DESCRIPTIONS

A-A-50177	-	Lens Paper

MILITARY SPECIFICATIONS

MIL-PRF-49040	-	Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter, MX-9644/UV
MIL-PRF-49367	-	Lens Assembly, Objective, 15mm AN/TVS-5
MIL-PRF-49368	-	Lens Assembly, Eyepiece, AN/PVS-4 and AN/TVS-5

(Copies of these documents are available online at http://quicksearch.dla.mil/)

2.2.2 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.i).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-850080	- Daylight Cover
SM-D-850100	- Night Vision Sight, Crew Served Weapon, AN/TVS-5
SM-D-850110	- Housing, Battery
SM-D-850120	- Eyeguard Assembly
SM-D-850200	- Case, Carrying and Storage
SM-D-850202	- Bag, Carrying
A3009873	- BA-3058/U ("AA") Battery Adapter

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2.3 Non-Government Publications The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2.i).

ANSI C18.1M	Batteries and Dry Cells Specifications
	Specification Sheet 20-0571-1988

(Copies of these documents are available on-line at http://ansi.org/)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Night Vision Sight, Crew Served Weapon, AN/TVS-5, herein referred to as the sight, is intended for use at night in aimed firing of crew served weapons. The sight consists of an Objective lens assembly (MIL-PRF-49367), an Eyepiece lens assembly (MIL-PRF-49368), an Eyeguard assembly (SM-D-850120), a BA-3058/U ("AA") Battery Adapter (A3009873), an Image intensifier assembly (MIL-PRF-49040), a Battery housing (SM-D-850110), a Daylight cover (SM-D-850080), Lens paper (SM-A-50177), a Case (SM-D-850200) and a Bag (SM-D-850202), which are major components of the sight (see 2.2.1 and 2.2.2).

3.2 Construction. The sight shall be constructed to assure compliance with the contract configuration baseline and in accordance with SM-D-850100 as applicable.

3.3 First Article Testing. When specified in the contract (6.2.g), the contractor shall furnish sights for first article testing (FAT) inspection and approval (see 4.2).

3.4 Material. The contractor shall select parts and materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible

3.5 Major components. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.f)

3.6 Performance Characteristics.

3.6.1 Eyeguard operation. With the eyeguard vertical and at -48°C, the eyeguard shall open (depress) when a 2000 ±10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100 ±10 grams. At +23°C, the eyeguard shall open (depress) when a 1600 ±10 gram weight is placed axially on the eye flange and shall close (expand) when the weight is reduced to 100 ±10 grams. At +23°C and with the eyeguard open (depressed) the minimum open position of the 2 leaves shall be 1.875 inches.

3.6.2 Sight definition. For conformance to the following performance parameters, the sight is defined as sight less case, bag and ancillary equipment.

3.6.3 Resolution. The minimum sight resolution shall be within one element number of the resolution of the image intensifier used in the sight at the time of

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testing while using a USAF 1951 Resolving Power Test Target with the target brightness adjusted to achieve best contrast.

3.6.4 Field of view. The sight shall have a minimum field of view of 156 milliradians.

3.6.5 Weight. The weight of the sight with batteries but without daylight cover and ancillary equipment shall not exceed 8.5 pounds.

3.6.6 Controls and adjustments. All controls and adjustments shall operate smoothly without binding throughout their complete range of movement at all operating temperatures specified herein.

3.6.6.1 Eyepiece diopter focus. With the diopter adjustment control set at -6 (+1/4, -0.0) to +2 diopter the image of the projected reticle shall be in sharp focus when viewed through a diopter scope of 3 power or greater.

3.6.6.2 Range focus. The sight shall provide a minimum focus adjustment range from a near focus of 25 meters to infinity. The dynamic torque to turn the range focus control shall be 8 ±7 inch-pounds.

3.6.6.3 Tube brightness control. The tube brightness control shall vary the gain of the image intensifier tube from OFF to maximum gain.

3.6.6.4 Reticle brightness control. Brightness of the reticle pattern shall be continuously variable over the range from OFF to full brightness. The pattern shall be visible and the numbers legible when viewing a solid white target having a brightness of 3.3×10^{-3} footlamberts minimum.

3.6.6.5 Reticle movement. Two external control knobs shall control the position of the reticle pattern viewed by the observer using the sight. One control shall move the reticle pattern in azimuth; another control shall move the reticle pattern in elevation. An adjustment made by turning one control shall not cause the reticle to deviate from a straight line of travel by more than 0.5 milliradian. Adjustment will be apparent to the operator by tactile indexing with audible clicks. Each click of the control knob shall displace the reticle pattern 0.25, ± 0.04 milliradian. Total movement of the reticle pattern shall be a minimum of 2.5° in any direction from the optical axis. The torque required to adjust the reticle position shall be between 0.5 and 4.0 inch-pounds. The reticle pattern shall remain in focus throughout the full range of movement of the range focus control. The reticle pattern shall not exhibit rotation greater than 1° from its proper orientation.

3.6.6.6 Battery adapter. The sight shall properly operate with the battery adapter attached and Type BA-3058/U batteries conforming to ANSI C18.1M Batteries and Dry Cells Specifications, Specification Sheet 20-0571-1988 inserted. The daylight cover may be installed during this test. Observe the sight for proper illumination.

3.7 Environmental Characteristics.

3.7.1 Temperature Extremes. The sight shall not be damaged (see 3.10.1) by storage in any temperature from -51°C to +68°C or by operation at any temperature from -51°C to +52°C. The sight shall not be damaged (see 3.10.1) after being subjected to repetitive temperature changes between +23°C and +68°C in a 5 minute period and between +23°C and -51°C in a 5 minute period. The transition from +68°C to -51°C and from -51°C to +68°C shall include a 2 hour soak at +23°C. Soak time at temperature extremes shall be sufficient for the entire sight to attain

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stabilization. Validation of operation shall be observance of a green glow at the eyepiece at operating temperature and, at low operating temperature, the ability to operate all controls with arctic gloves or equivalent

3.7.2 Vibration. The sight shall not be damaged (see 3.10.1) after being subjected to simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion) with the frequency being varied between 5 and 55 Hertz (Hz). Operation of the sight during vibration shall not be required. Vibration shall be applied in 3 mutually perpendicular planes, one of which is perpendicular to the optical axis. The duration of vibration shall be 5 minutes in each plane. The test after vibration shall be resolution (3.6.3).

3.7.3 High intensity shock. The sight shall not be damaged (see 3.10.1) by a sequence of 3 shocks applied in each direction along each of 3 mutually perpendicular axes (horizontal, vertical, and optical axis when the reticle pattern is erect) with the sight in the non-operating mode. The shocks shall be half sine pulses and shall have a time duration of 4 milliseconds, ±5 percent. Shock pulses applied along the axes horizontal and perpendicular to the optical axis shall have a peak amplitude of 50 g's ±15 g's (see 3.10.3). Shock pulses applied along the optical axis shall have a peak amplitude of 100 ±15 g's (see 3.10.3).

3.7.4 Altitude. The sight shall not be damaged (see 3.10.1) when operated at a pressure equivalent to 10,000 feet altitude above sea level. The sight shall be operated for a period of 3 hours. Validation of operation shall be observance of a green glow at the eyepiece.

3.7.5 Humidity. The sight shall not be damaged (see 3.10.1) after being operated in an atmosphere with a relative humidity of 95 ±5 percent over a temperature range of +20°C to +52°C and shall operate after storage up to +68°C minimum with 95 ±5 percent relative humidity.

3.7.6 Immersion. The sight shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The sight shall be pre-conditioned by exposure to temperature of +45°C ±3°C for a period of 2 hours prior to being immersed in the fresh water at a temperature of +18°C ±5°C. The sight shall be inspected, without opening, for evidence of leakage.

3.7.7 Interchangeability. The objective lens assembly and the eyepiece lens assembly shall be physically interchangeable between sight assemblies without degradation of the resolution (see 3.6.3), eyepiece diopter focus (see 3.6.6.1), and range focus (see 3.6.6.2) requirements. These interchanges shall be made without any modification or adjustment beyond that necessary during replacement of any of the parts or assemblies involved.

3.8 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract.

3.9 Workmanship. Standards of workmanship shall be such that the sight will meet all requirements of this specification and any referenced specifications or drawings.

3.10 Technical interpretations. The following technical interpretations, when referenced in section 3 and 4, are mandatory for this specification.

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3.10.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; leakage or condensation of moisture within the sight; any degradation in input or output characteristics; and any failure of the control knobs to perform their function.

3.10.2 Zero. A known and specified relationship between the sight aiming point and the strike of the bullet.

3.10.3 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

3.10.4 Room temperature. Room temperature is defined as +23°C, ±10°C.

3.10.5 Environmental temperature. Environmental chamber temperatures shall be controlled to within ±5°C of the specified temperature unless otherwise specified.

3.10.6 Illumination levels shall be ±10% of the light level specified.

3.10.7 Tolerance on 2854K color temperature shall be ±200K.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article testing (4.2).
b. Conformance inspection (4.3).

4.2 First Article Testing. When specified in the contract or purchase order (see 6.2.d), first article testing shall be performed by the contractor. The quantity of initial production test samples shall be as specified in the contract or purchase order and shall be selected from the first production lot.

4.2.1 FAT Test. Each sight in the FAT lot shall be subjected to the inspections of Table I. Samples selected from the FAT lot shall be subjected to the inspections of Table II. Sample size shall be as specified in the contract or purchase order (see 6.2.g). Failure of any inspection shall be cause for rejection of that sight and may be cause for failure of the FAT. Inspections may be conducted in any order except for Immersion shall follow temperature extremes and temperature shock.

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TABLE I. First article testing unit test.

Inspection	Requirement Paragraph
Resolution	3.6.3
Field of View	3.6.4
Eyepiece focus	3.6.6.1
Range focus	3.6.6.2
Tube brightness	3.6.6.3
Reticle brightness	3.6.6.4
Reticle movement	3.6.6.5

Table II First article testing sample test.

Inspection	Requirement Paragraph
Eyeguard Operation	3.6.1
Temperature extremes	3.7.1
Vibration	3.7.2
High intensity shock	3.7.3
Altitude	3.7.4
Humidity	3.7.5
Immersion	3.7.6
Interchangeability	3.7.7
Weight	3.6.5

4.2.2 Disposition of FAT samples. After successful completion of the FAT, the FAT sample units shall be refurbished and delivered in accordance with the contract or purchase order (see 6.2.g).

4.3 Conformance inspection. Requirements for conformance inspection shall be as specified in the contract or purchase order (see 6.2.f).

4.4 Operational test. The test of the sight required during environmental testing shall consist of power up of the sight and observance of the green glow at the eyepiece.

5. PACKAGING

5.1 Packaging requirements. Packaging and preservation must be best commercial practice unless otherwise specified in the contract or purchase order (see 6.2.j).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AN/TVS-5 is intended for use at night for aimed firing of crew served weapons.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Which of the following components are to be purchased and the quantities involved:
    M2 mounting bracket per SM-D-850220
    M40 mounting bracket per SM-D-850230
    M139, M80, tripod mounting bracket per SM-D-850240
    Bag, cloth, laminated vinyl-nylon SM-D-850202-1
    Bag, cloth nylon SM-D-850202-2
    Reticle M2, M85 per drawing SM-D-850490-1
    Reticle M40 per drawing SM-D-850490-2
    Reticle M139 per drawing SM-D-850490-6
    Reticle M60 per drawing SM-D-850490-8
    Batteries, BA 5567/U, BA-3058/U
c. If the image intensifier tube per MIL-PRF-49040 is to be purchased.
    (1) To obtain the sight with Image Intensifier Tube, specify the use of drawing SM-D-850100-1.
    (2) To obtain the sight without Image Intensifier Tube, specify the use of drawing SM-D-850100-2.
d. Statement specifying that FAT testing is, or is not, required and sampling requirement for FAT.
e. Necessary actions by the contractor in the event of a lot test failure.
f. Statement identifying that the contractor must submit his proposal for conformance inspection to include the specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
g. Quantity and schedule for FAT testing for all line items and disposition of FAT samples.
h. Prior to packing and packaging, each sight must be examined to assure absence of batteries.
i. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).
j. Packaging and preservation (see section 5).

6.3 Definitions. See 3.10.

6.4 Subject term keyword listing.
Image Intensification

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6.5 Changes from previous issue. Marginal notations are used in this revision to identify changes with respect to the previous issue. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue due to the extensive changes.

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